Britney T. Forsyth

New York, NY | britneyforsyth8@gmail.com | (626) 342-6231 | LinkedIn

EDUCATION

Columbia University | New York, NY — Sep 2021 – Dec 2022
MS Biomedical Engineering – Concentration in Bioinformatics

Pennsylvania State University | University Park, PA — Aug 2017 – May 2021
BS Biomedical Engineering – Concentration in Biochemistry
BS Mathematics – Concentration in Statistics

RESEARCH AND WORK EXPERIENCE

Bioinformatics Software Engineer | New York, NY — Jan 2023 - Present
Memorial Sloan Kettering Cancer Center
Exploring the Role of ZFP36L2 in Tumor Plasticity and Cell State Transitions during Colorectal Cancer

- Developed computational pipelines for preprocessing and quality control of scRNA-seq data with Python and R on HPC systems
- Applied machine learning techniques, data visualization strategies, and interpreted genomic data to identify gene expression patterns associated with tumor progression
- Collaborated closely with interdisciplinary teams and effectively communicated research outcomes in team meetings and conferences

Machine Learning Researcher | New York, NY — Sep 2021 – Dec 2022
Columbia University, Department of Biomedical Engineering
General Adversarial Networks for Predicting Wound Healing Trajectories Using scRNA-seq

- Performed computational analysis of scRNA-seq data with R, Python, SQL, and Bash scripting on HPC systems
- Generated a GANs network for augmentation of scRNA-seq data to understand wound healing over time
- Assisted with transcriptional analysis of spheroid culture via RNA isolation, rt-PCR, and gel electrophoresis
- Communicated research progress and results at scientific meetings of 7+ team members consisting of engineers, lab technicians, and physicians

Graduate Research Assistant | New York, NY — Jan 2023 – May 2023
Columbia University, Department of Biomedical Engineering | Mentor: Dr. Kaveri A. Thakoor
Predicting Clinician Fixations on Glaucoma OCT Reports via CNN-Based Saliency Prediction Methods

- Developed fixation heatmaps from eye-tracking data using a Pupil Labs Core eye-tracker, identifying regions of interest on OCT reports
- Trained a TranSalNet model with traditional saliency mapping, achieving a correlation coefficient (CC) value of 0.208, NSS value of 0.8172, KLD value of 2.573, and SSIM of 0.169
- Suggested future research directions to enhance model accuracy, including increasing data collection, improving data quality, and modifying model architecture

Graduate Research Assistant | New York, NY — Sep 2021 – Dec 2022
Columbia University, Department of Biomedical Engineering | Mentor: Dr. Jia Guo
Cell Segmentation and Tracking with Residual Attention U-Net

- Generated a dataset of 300+ cell microscopy masks with ImageJ for algorithm training
- Developed a deep learning model for segmentation and tracking of fluorescent cell microscopy images with Python and Python libraries, such as PyTorch, NumPy, Pandas, and scikit
- Achieved 97% segmentation accuracy on testing dataset using a novel convolutional neural network (CNN) with residual blocks and attention mechanisms
- Communicated research progress and results at scientific meetings of 10+ team members consisting of engineers, lab technicians, and physicians

Research Intern | Bethesda, MD May 2021 – Aug 2021
National Institutes of Health | Mentors: Dr. Lawrence Latour, Dr. Marie Luby
Deep Learning Algorithm for Classification of Hemorrhagic Transformation in MRI

- Created and implemented a Python and MATLAB-based algorithm for classification and severity prediction of hemorrhagic transformation with human brain MRI images
- Collaborated with physicians, engineers, and lab technicians to compile a clinical stroke patient dataset of 100+ MRI images for algorithm training
- Achieved testing accuracy of 80% on testing dataset with novel implementation of a 3D CNN in TensorFlow

Undergraduate Research Assistant | University Park, PA Dec 2017 – May 2021
Pennsylvania State University, Department of Biomedical Engineering | Mentor: Dr. Pak K. Wong
A Rapid Antimicrobial Susceptibility Workflow for Diagnosis of Bloodstream Infections

- Designed, simulated, and tested microfluidic lab-on-chip devices with CAD and 3D printing
- Performed cell culture of bloodborne pathogens, including inoculation and maintenance of reagents in sterile conditions
- Conducted molecular analysis with Gram staining and PCR for identification of bacterial species and prepared cells for fluorescent imaging via staining with fluorescent dyes
- Optimized workflow for diagnosis of bloodstream infections to isolate blood-borne pathogens within 5-6 hours. Supervised and mentored 3 undergraduate students to coordinate tasks for fabrication and testing of microfluidic devices

Research Intern | Bethesda, MD May 2020 – Aug 2020
National Institutes of Health

- COVID cancelled

Research Intern | Bethesda, MD May 2019 – Aug 2019
National Institutes of Health | Mentor: Dr. Carlo Pierpaoli
TORTOISE: A Python-Based Program for Correction of MRI Images

- Constructed image-quality-enhancement algorithms to improve 3D image quality, image reconstruction, procession, and analysis of 100+ diffusion-weighted MRI (DW-MRI) images in Python and C
- Collaborated and communicated with 7+ team of engineers, scientists, and physicians to determine proper correction of DW-MRI images for release of TORTOISE software
- Achieved better image quality, faster imaging speed and higher robustness of DW-MRI image quality for improved image analysis

PUBLICATIONS

Zang, M., Mukund, P., **Forsyth, B.**, Laine, A. F., & Thakoor, K. A. (2024). Predicting Clinician Fixations on Glaucoma OCT Reports via CNN-Based Saliency Prediction Methods. *IEEE open journal of engineering in medicine and biology*, *5*, 191–197. https://doi.org/10.1109/OJEMB.2024.3367492

Lee, P. Y., Wei, H. J., Pouliopoulos, A. N., **Forsyth, B. T.**, Yang, Y., Zhang, C., Laine, A. F., Konofagou, E. E., Wu, C. C., & Guo, J. (2023). Deep Learning Enables Reduced Gadolinium Dose for Contrast-Enhanced Blood-Brain Barrier Opening. *ArXiv*, arXiv:2301.07248v1.

Zhu, N., Liu, C., **Forsyth, B.**, Singer, Z. S., Laine, A. F., Danino, T., & Guo, J. (2022). Segmentation with Residual Attention U-Net and an Edge-Enhancement Approach Preserves Cell Shape Features. *Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE Engineering in Medicine and Biology Society. Annual International Conference*, *2022*, 2115–2118. https://doi.org/10.1109/EMBC48229.2022.9871026

Forsyth, **B**. et al. "A Rapid Single-Cell Antimicrobial Susceptibility Testing Workflow for Bloodstream Infections." *Biosensors* vol. 11,8 288. 22 Aug. 2021, doi:10.3390/bios11080288

CONFERENCE PRESENTATIONS

Forsyth, B., Lallo, M., Jiang, Q., Ganesh, K., Chan, J. Understanding the Role of ISC-like Genes in Tumor Plasticity and Cell State Transitions in Colorectal Cancer. Presented at: Memorial Sloan Kettering's Geoffrey Beene Conference. May 2024. New York, NY.

Forsyth, B., Woappi, Y. Predicting Wound Healing Trajectories of Single-Cell RNA-Sequencing Data with Generative Modeling. Poster presented at: Mount Sinai NY Skin Club Conference. Dec 2023. New York, NY.

Zhu, N., Liu, C., **Forsyth**, **B**., Singer, Z., Danino, T., Laine, A., Guo, J. Segmentation with Residual Attention U-Net and An Edge-Enhancement Approach Preserves Cell Shape Features. Poster presented at: 44th Annual International Conference of the IEEE Engineering in Medicine and Biology Society. July 2022. Glasgow, Scotland.

Forsyth, **B**., Torab, P., Lee, J. H., Wong, P. K. Developing A Diagnostic Workflow for Sepsis Detection: A Research-Based Experience in Penn State's SEN Grant Program. Poster presented at: International Conference of University Service-Learning at University of Las Palmas de Gran Canaria. July 2021. Canaria, Spain.

Forsyth, **B**., Jackson, S., Latour, L., Luby, M. A Deep Learning Algorithm for Classification of Hemorrhagic Transformation in MRI. Poster presented at: National Institutes of Health Summer Poster Day. August 2021. Bethesda, MD.

Forsyth, **B**., Torab, P., Lee, J. H., Wong, P. K. A Rapid Antimicrobial Susceptibility Testing Workflow for Characterization of Bloodstream Pathogens. Poster presented at: Penn State Undergraduate Exhibition. May 2021. University Park, PA.

Forsyth, **B**., McIntyre, R., Wong, P. K. A Rapid Antimicrobial Susceptibility Testing Protocol for Sepsis Diagnosis. Poster presented at: Penn State Undergraduate Exhibition. May 2020. University Park, PA.

Forsyth, **B**., Thai, A., Pierpaoli, C., Irfanoglu, M. O. Motion and Eddy-Current Distortion Correction of Diffusion MRI with Python: TORTOISE in DIPY. Poster presented at: BMES Annual Meeting. October 2020. San Diego, CA.

Forsyth, **B**., Thai, A., Pierpaoli, C., Irfanoglu, M. O. Motion and Eddy-Current Distortion Correction of Diffusion MRI with Python: TORTOISE in DIPY. Poster presented at: National Institutes of Health Summer Poster Day. August 2019. Bethesda, MD.

Torab, P., **Forsyth**, **B**., Malcolm, T., Lu, Y., Wong, P. K. Rapid Bacterial Enrichment for Sepsis Detection and Antimicrobial Susceptibility Testing. Poster presented at: BMES Annual Meeting. October 2019. Philadelphia, PA.

AWARDS, GRANTS, AND FELLOWSHIPS

2022 NSF Fellowship for 19th International Summer School on BIO-X
2021 Penn State SEN Remote Innovation Grant
2020 College of Engineering Equity REU Scholarship
2018 Collegiate Laws of Life Essay Winner
2017 Provost Scholarship

TEACHING EXPERIENCE

Organic Chemistry and Biology Tutor | Remote May 2019 – June 2023
Outside the Box Academy

- Tutored 5+ students in Organic Chemistry I, Calculus, Geometry, Physiology, and Biology
- Generated lesson materials, lecture presentations, and homework assignments for 2+ hour sessions
- Collaborated with parents and tutors to generate student-specific lesson plans and identify content for review

Mathematics Grader and Exam Proctor | University Park, PA Dec 2017 – May 2020
Pennsylvania State University, Department of Mathematics

- Graded weekly homework and quizzes for undergraduate courses of 50+ students in differential equations, matrices, and statistics

- Observed students for any violations in academic integrity for 100+ students in undergraduate math courses during exams
- Resolved grading disputes and liaised with professors, students, and mathematics department to generate feedback about course material

SKILLS

Programming: Python, R, SQL, MATLAB, Bash, HTML/CSS
Software: Microsoft Office (Excel, PowerPoint, Word), Adobe (Photoshop, Illustrator, Premiere Pro), R Studio, Anaconda Navigator, Jupyter Notebook, GitHub, SPSS, SAS, COMSOL, CAD, ImageJ
Laboratory: Cell Culture, rt-PCR, Gel Electrophoresis, RNA/DNA Isolation, Fluorescent Microscopy

COMMUNITY INVOLVEMENT

THON Donor and Alumni Relations Fundraising Specialist | University Park, PA Oct 2017 – May 2021

- Established and maintained relationships with 1000+ volume of potential, current, and past donors
- Served as the liaison between THON's 16,000 student volunteers and 25,000 annual alumni networks
- Fostered relationships between donors, alumni, and volunteers through presentations, THONvelope distribution, and guided tours of the arena during THON Weekend

REFERENCES

Dr. Pak K Wong
Pennsylvania State University
Professor of Biomedical Engineering and Mechanical Engineering
516 Chemical and Biomedical Engineering Building, University Park, PA 16802
pxw28@psu.edu

Dr. Joseph Chan
Memorial Sloan Kettering Cancer Center
Assistant Professor Human Oncology and Pathogenesis Program
321 E 61st St, New York, NY 10065
chanj3@mskcc.org

Dr. Marie Luby
University of Connecticut
Adjunct Faculty of Biomedical Engineering
260 Glenbrook Road, Storrs, CT, 06269
Marie.luby@uconn.edu